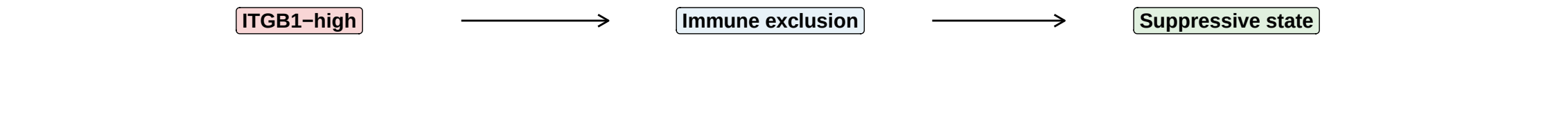
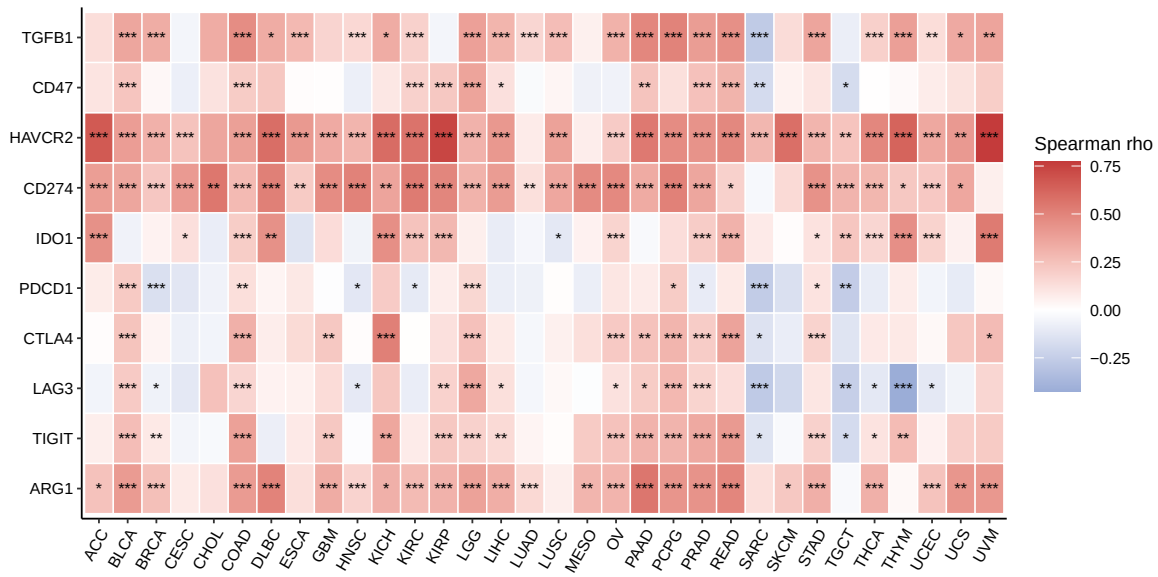
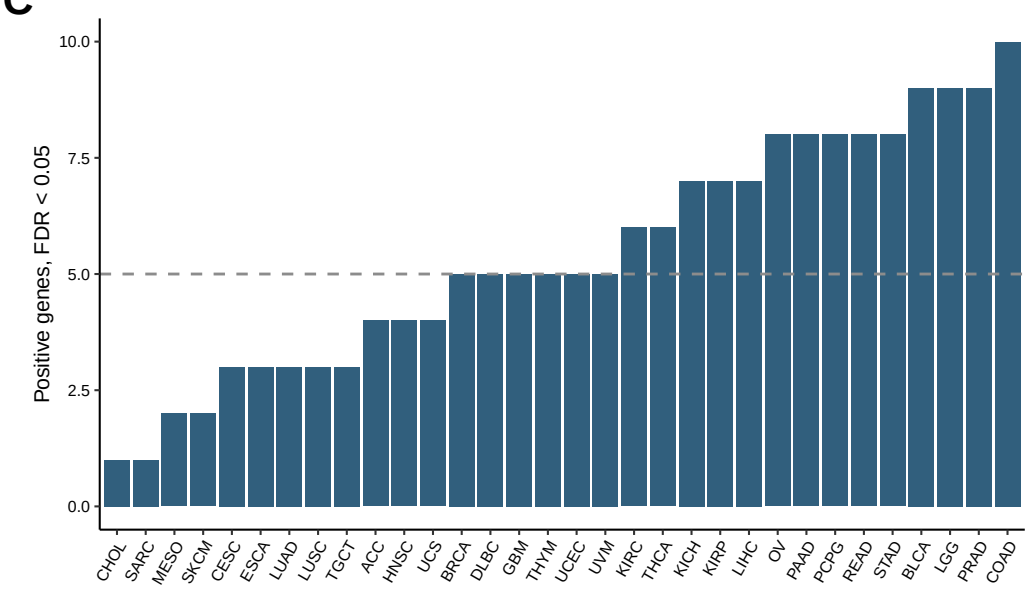
A


ITGB1−high
Immune exclusion
Suppressive state

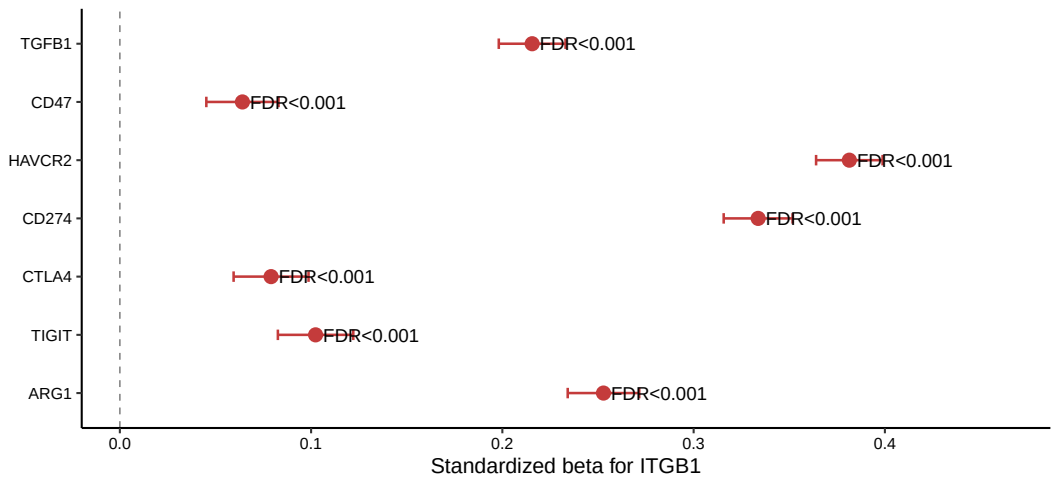
B


TGFB1
CD47
HAVCR2
CD274
IDO1
PDCD1
CTLA4
LAG3
TIGIT
ARG1
ACC BLCA BRCA CESC CHOL COAD DLBC ESCA GBM HNSC KICH KIRC KIRP LGG LIHC LUAD LUSC MESO OV PAAD PCPG PRAD READ SARC SKCM STAD TGCT THCA THYM UCEC UCS UVM
Spearman rho
0.75
0.50
0.25
0.00
−0.25

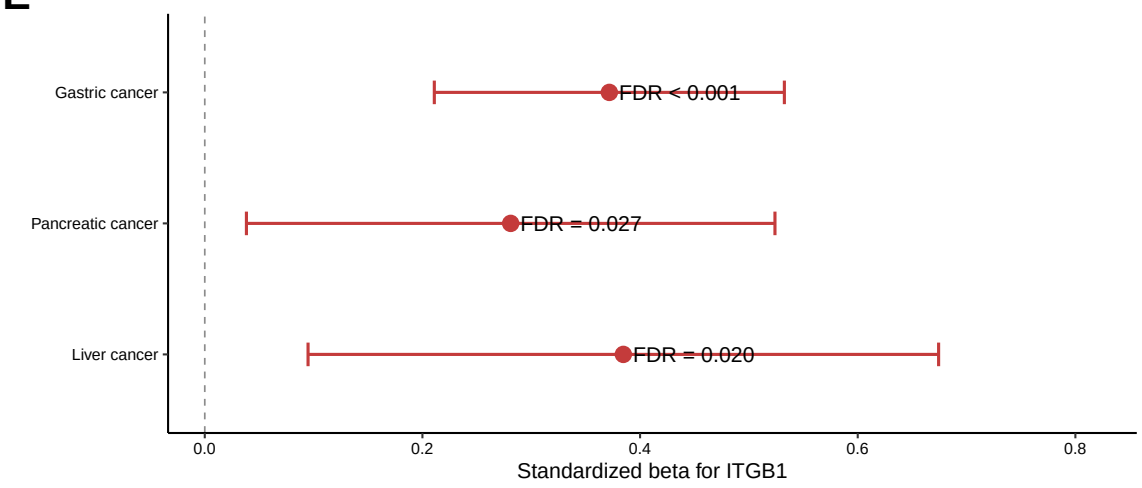
C


Positive genes, FDR < 0.05
10.0
7.5
5.0
2.5
0.0
CHOL SARC MESO SKCM CESC ESCA LUAD LUSC TGCT ACC HNSC UCS BRCA DLBC GBM THYM UCEC UVM KIRC THCA KICH KIRP LIHC OV PAAD PCPG READ STAD BLCA LGG PRAD COAD

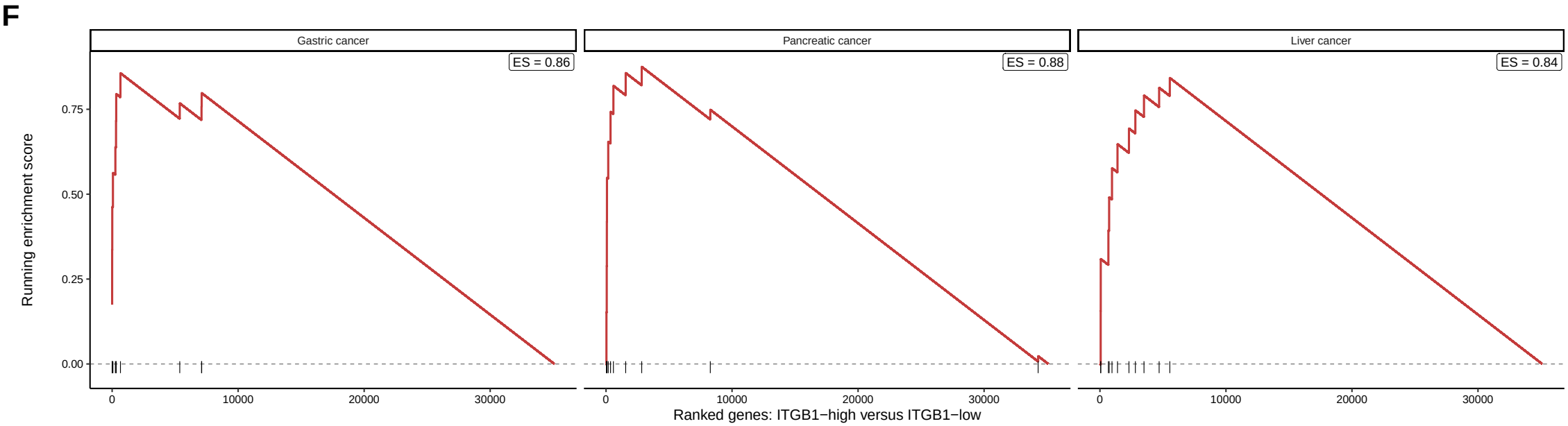
D


TGFB1 FDR<0.001
CD47 FDR<0.001
HAVCR2 FDR<0.001
CD274 FDR<0.001
CTLA4 FDR<0.001
TIGIT FDR<0.001
ARG1 FDR<0.001
0.0 0.1 0.2 0.3 0.4
Standardized beta for ITGB1

E


Gastric cancer FDR < 0.001
Pancreatic cancer FDR = 0.027
Liver cancer FDR = 0.020
0.0 0.2 0.4 0.6 0.8
Standardized beta for ITGB1

F


Gastric cancer
ES = 0.86
Pancreatic cancer
ES = 0.88
Liver cancer
ES = 0.84
Running enrichment score
0.75
0.50
0.25
0.00
0 10000 20000 30000
Ranked genes: ITGB1−high versus ITGB1−low